$$\lim_{x \to \pm\infty} f(x) = c \quad \lim_{x \to x^0} f(x) = \pm\infty \quad m = \frac{\lim_{x \to \infty} f(x)}{x}, \; q = \lim_{x \to \infty} f(x) - mx; \lim_{x \to \infty} f(x) - mx - q = 0. \; y = mx + q$$

$$y = f'(x_0) \cdot x + f(x_0) - (f'(x_0) \cdot x_0) \qquad g'(x) = \frac{1}{f'(g(x))} \qquad x - x_0 = g'(x)(y - y_0)$$

$$(\cos(x))^2 + (\sin(x))^2 = 1 \qquad (tanx)^2 + 1 = \frac{1}{(cosx)^2}$$

$$\sin(-\alpha) = -\sin(\alpha) \qquad \cos(-\alpha) = -\cos(\alpha) \qquad \tan(-\alpha) = -\tan(\alpha)$$

$$\sin\left(\alpha + \frac{\pi}{2}\right) = \cos\alpha \qquad \cos\left(\alpha + \frac{\pi}{2}\right) = -\sin\alpha \qquad \sin\left(\alpha - \frac{\pi}{2}\right) = -\cos\alpha \qquad \cos\left(\alpha - \frac{\pi}{2}\right) = \sin\alpha$$

$$\sin(\alpha \pm \pi) = -\sin\alpha \qquad \cos(\alpha \pm \pi) = -\cos\alpha$$

$$\sin\left(\alpha + \frac{3\pi}{2}\right) = -\cos\alpha \quad \cos\left(\alpha + \frac{3\pi}{2}\right) = \sin\alpha \quad \sin\left(\alpha - \frac{3\pi}{2}\right) = \cos\alpha \quad \cos\left(\alpha - \frac{3\pi}{2}\right) = -\sin\alpha$$

$$\tan\left(\alpha \pm \frac{\pi}{2}\right) = -\cot\alpha \qquad \cot\left(\alpha \pm \frac{\pi}{2}\right) = -\tan\alpha \qquad \tan(\alpha \pm \pi) = \tan\alpha \qquad \cot(\alpha \pm \pi) = -\cot\alpha$$

$$\sin(x \pm y) = \sin(x)\cos(y) \pm \cos(x)\sin(y) \quad \cos(x \pm y) = \cos(x)\cos(y) \mp \sin(x)\sin(y)$$

$$\tan(\alpha \pm \beta) = \frac{\tan(\alpha) \pm \tan(\beta)}{1 + \tan(\alpha) \cdot \tan(\beta)} \; \sin\left(\frac{\alpha}{2}\right) = \pm\sqrt{\square} \; \cos\left(\frac{\alpha}{2}\right) = \pm\sqrt{\square} \; \tan\left(\frac{\alpha}{2}\right) = \pm\sqrt{\square}$$

$$\sin(2x) = 2 \; \sin(x) \; \cos(x) \qquad \cos(2x) = (\cos(x))^2 - (\sin(x))^2 = 2 \; (\cos(x))^2 - 1 = 1 - 2 \; (\sin(x))^2$$

$$(sinx)^2 = \frac{1 - \cos(2x)}{2} \qquad (cosx)^2 = \frac{1 + \cos(2x)}{2}$$

$$\sinh(x) = \frac{e^x - e^{-x}}{2} \qquad \cosh(x) = \frac{e^x + e^{-x}}{2} \qquad \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \qquad (\cosh(x))^2 - (\sinh(x))^2 = 1 \qquad 1 - (tanhx)^2 = \frac{1}{(coshx)^2}$$

$$\sinh(2x) = 2 \; \sinh(x) \; \cosh(x) \qquad \cosh(2x) = (\cosh(x))^2 + (\sinh(x))^2 = 2 \; (\cosh(x))^2 - 1 = 1 + 2 \; (\sinh(x))^2$$

$$\sinh(x \pm y) = \sinh(x)\cosh(y) \pm \cosh(x)\sinh(y) \qquad \cosh(x \pm y) = \cosh(x)\cosh(y) \pm \sinh(x)\sinh(y)$$

$$(sinhx)^2 = \frac{\cosh(2x) - 1}{2} \quad (coshx)^2 = \frac{\cosh(2x) + 1}{2} \qquad e^x = coshx + sinhx \qquad e^{-x} = coshx - sinhx \qquad 1 - (\tanh(x))^2 = \frac{1}{(\cosh(x))^2}$$

$$sett \; \sinh(\alpha) = \ln(\alpha + \sqrt{\square}) \qquad sett \; \cosh(\alpha) = \ln(\alpha + \sqrt{\square}) \qquad sett \; \tanh(\alpha) = \frac{1}{2}\ln\left(\frac{1+\alpha}{1-\alpha}\right) = \ln\left(\frac{\sqrt{\square}}{\square}\right)$$

$$sett \; sech(\alpha) = \ln\left(\frac{1 + \sqrt{\square}}{\square}\right) \qquad sett \; csch(\alpha) = \ln\left(\frac{1 \pm \sqrt{\square}}{\square}\right) \qquad sett \; \coth(\alpha) = \ln\left(\frac{\sqrt{\square}}{\square}\right)$$

$$+\infty - \infty \quad 0 \cdot (\pm\infty) \qquad \frac{\pm\infty}{\pm\infty} \qquad \frac{0}{0} \qquad 0^0 \qquad 1^{\pm\infty} \qquad (\pm\infty)^0$$

$$(costante)' = 0 \quad (x^\alpha)' = \alpha \cdot x^{\alpha - 1} \quad (e^x)' = e^x \qquad (a^x)' = a^x \cdot \log(a) \qquad (\log(x))' = \frac{1}{x} \qquad (\sin(x))' = \cos(x)$$

$$(\cos(x))' = -\sin(x) \qquad (\tan(x))' = \frac{1}{(\cos(x))^2} = 1 + (\tan(x))^2 \qquad (\cot(x))' = -\frac{1}{(\sin(x))^2} \qquad (\arctan(x))' = \frac{1}{1 + x^2}$$

$$(\operatorname{arccot}(x))' = -\frac{1}{1 + x^2} \qquad (\arcsin(x))' = \frac{1}{\sqrt{\square}} \qquad (\arccos(x))' = -\frac{1}{\sqrt{\square}} \qquad (\sinh(x))' = \left(\frac{e^x - e^{-x}}{2}\right)' = \frac{e^x + e^{-x}}{2} = \cosh(x)$$

$$(\cosh(x))' = \left(\frac{e^x + e^{-x}}{2}\right) = \frac{e^x - e^{-x}}{2} = \sinh(x) \qquad (\tanh(x))' = \left(\frac{e^x - e^{-x}}{e^x + e^{-x}}\right)' = \frac{4 \; e^{2x}}{(e^{2x} - 1)^2} = \frac{1}{(\cosh(x))^2} = 1 - (\tanh(x))^2$$

$$(\coth(x))' = \frac{1}{(\sinh(x))^2}$$

$$(f \cdot g)' = f'g + fg' \qquad \left(\frac{1}{f}\right)' = -\frac{f'}{f^2} \qquad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2} \qquad (f(g(x)))' = f'(g(x)) \cdot g'(x)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots \quad \sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} + \ldots \quad \cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} + \ldots$$

$$\tan(x) = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \frac{17}{315}x^7 + \ldots \quad \sinh(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \frac{x^9}{9!} + \ldots \quad \cosh(x) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \frac{x^8}{8!} + \ldots$$

$$\tanh(x) = x - \frac{x^3}{3} + \frac{2}{15}x^5 - \frac{17}{315}x^7 + \ldots$$

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} + \ldots \qquad \arctan(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \ldots$$

$$\tanh(x) = x - \frac{x^3}{3} + \frac{2}{15}x^5 - \frac{17}{315}x^7 + \ldots \quad arctanh(x) = x + \frac{x^3}{3} + \frac{x^5}{5} + \frac{x^7}{7} + \ldots \qquad arcos(x) = \frac{\pi}{2} - x - \frac{x^3}{6} - \frac{3}{40}x^5$$

$$\arcsin(x) = x + \frac{x^3}{6} + \frac{3}{40}x^5 \qquad a^x = 1 + x\ lna + \frac{x^2}{2}\ln^2 a + \frac{x^3}{3!}\ln^3 a$$

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!}x^3 + \frac{\alpha(\alpha-1)(\alpha-2)(\alpha-3)}{4!}x^4 + \ldots$$

$$f(x) \in \textit{intorno di } x_0 = f(x_0) + \frac{f'(x_0)}{1!}(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 \ldots$$

$$\frac{\lim_{x\to 0}\sin(x)}{x} = 1 \qquad \lim_{x\to\pm\infty}\left(1+\frac{a}{x}\right)^x = e^a \qquad \frac{\lim_{x\to 0}1-\cos(x)}{x^2} = \frac{1}{2} \qquad \frac{\lim_{x\to 0}e^x - 1}{x} = 1 \qquad \frac{\lim_{x\to 0}\log_a(1+x)}{x} = \frac{1}{\log\ a}$$

$$\frac{\lim_{x\to 0} tg(x)}{x} = 1 \qquad \frac{\lim_{x\to 0} arctg(x)}{x} = 1 = \frac{\lim_{x\to 0}\arcsin(x)}{x} \qquad \frac{\lim_{x\to 0}a^x - 1}{x} = \log(a) \qquad \lim_{x\to 0^+} x\ \log(x) = 0 \qquad \lim_{x\to 0}(1+x)^{\frac{1}{x}} = e$$

$$\frac{\lim_{x\to 0}(1+x)^a - 1}{x} = a$$

$$n^n,\ n!,\ a^n,\ n^b,\ \log\ n$$

$$\int_{\square} x^\alpha = \frac{x^{\alpha+1}}{\alpha+1} \qquad \int_{\square}\frac{1}{x} = \log|x| \qquad \int_{\square}\sin(x) = -\cos(x) \qquad \int_{\square}\cos(x) = \sin(x) \qquad \int_{\square}\sinh(x) = \cosh(x) \qquad \int_{\square}\cosh(x) = \sinh(x)$$

$$\int_{\square} e^x = e^x \qquad \int_{\square} a^x = \frac{a^x}{\ln(a)} \qquad \int_{\square}\frac{1}{1+x^2} = \arctan(x) \qquad \int_{\square} -\frac{1}{1+x^2} = \operatorname{arccot}(x) \qquad \int_{\square}\frac{1}{1-x^2} = arctanh(x) \qquad \int_{\square}\frac{1}{\sqrt{\square}} \quad \int_{\square}\frac{1}{\sqrt{\square}} \quad \int_{\square} -\frac{1}{\sqrt{\square}}$$

$$\int_{\square}\frac{1}{\sqrt{\square}} \qquad \int_{\square}\frac{1}{(\cos(x))^2} = \int_{\square} 1 + (\tan(x))^2 = \tan(x)$$

$$\int_{\square}\frac{1}{(\cosh(x))^2} = \int_{\square} 1 - (\tanh(x))^2 = \tanh(x) \qquad \int_{\square}\frac{1}{\sqrt{\square}} \qquad \int_{\square} -\frac{1}{(\sin(x))^2} = cotan(x)$$

$$\int_{\square}\frac{k}{a\,x^2+bx+c}\,dx = 2k\,\frac{\arctan\left(\frac{2ax+b}{\sqrt{\square}}\right)}{\square}$$

Lunghezza curva $\int_a^b \sqrt{\square}$ Volume sotteso curva $\pi\int_a^b f(x)^2$ Volume asse y $2\pi\int_a^b x\cdot f(x)$

$$t = \tan\left(\frac{\alpha}{2}\right) \qquad \sin\alpha = \frac{2t}{1+t^2} \qquad \cos\alpha = \frac{1-t^2}{1+t^2}$$

$$\int_0^\infty \frac{1}{x^\alpha \ln(x)^\beta}$$

Problema in 0 $\alpha < 1$ Problema in ∞ $\alpha > 1$ oppure $\alpha = 1$ e $\beta > 1$

Se $f'(x_0) = 0$, trovo la prima $f^{(k)}(x_0) \neq 0$.

$k\ pari\ <0 \Rightarrow Max\ locale$ $k\ pari\ >0 \Rightarrow Min\ locale$ $k\ dispari\ <0 \Rightarrow Flesso\ disc$ $k\ dispari\ >0 \Rightarrow Flesso\ asc$

$$y'(t) + p(t)\,y(t) = q(t) \Rightarrow y(t) = e^{-P(t)}\left(\int_{\square} q(t)\,e^{+P(t)}\,dt + c\right)$$

Secondo ordine omogenee $ay''+by'+cy=0$

$\Delta>0 \Rightarrow y(t)=c_1 e^{\lambda_1 t}+c_2 e^{\lambda_2 t}$ $\quad \Delta=0 \Rightarrow y(t)=c_1 e^{\lambda t}+c_2 t e^{\lambda t}$ $\quad \Delta<0 \Rightarrow y(t)=c_1 e^{\alpha t}\sin(\beta t)+c_2 e^{\alpha t}\cos(\beta t)$ $\quad \alpha \pm \beta i$

Secondo ordine non omogenee $ay''+by'+cy=f(t)$

$f(t)=e^{\alpha t} \Rightarrow \underline{u}(t)=\lambda(t)e^{\alpha t}$ $\quad f(t)=\cos(\alpha t) \Rightarrow \underline{u}(t)=t(\lambda \sin(\alpha t)+\mu\cos(\alpha t))$

$f(t)=$ *polinomio grado* $s \Rightarrow \underline{u}(t)=t(at^s+bt^{s-1}+...)$